One-Variable Statistics, Distributions & Margin of Error

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A data set has values 4, 7, 9, 11, 14. What is the median?

2 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the data set: $84, 37, 93, 17, 57, 98$?

3 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the data set: $50, 23, 12, 96, 77, 47$?

4 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $17, 46, 37, 12, 20, 48$?

5 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $30, 35, 35, 16, 49$?

6 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the data set: $31, 65, 66, 28, 29$?

7 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the data set: $62, 67, 97, 6, 1$?

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The table shows the number of pets owned by each student in a class. What is the mean number of pets per student?

mber of p	0	1	2	3	4
ber of stu	1	1	1	5	2

9 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a survey, 40% of 200 students prefer math. How many students prefer math?

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10 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $20, 7, 15, 14, 14$?

11 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mode of the data set: $44, 16, 10, 44, 45, 9, 12$?

12 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mode of the data set: $33, 34, 27, 3, 4, 34$?

13 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The table shows test scores for two classes. What is the positive difference between the medians?

14 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The mean of 4 numbers is 36. When one more number, 31, is added to the set, what is the new mean of all 5 numbers?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The mean of 5 numbers is 12. Four of them are 8, 10, 14, 16. What is the fifth?

16 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The range of the data set {3, 7, 2, 9, 5, 8} is

17 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $1, 28, 23, 49, 45, 52$?

18 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of the data set: $4, 17, 53, 5, 64$?

19 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the data set: $7, 40, 31, 18$?

20 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of the data set: $45, 41, 59, 13, 65, 54, 68$?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mode of the data set: $20, 19, 45, 43, 20, 12$?

a. 20

b. 27

c. 28

d. 11

22 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of the following list of numbers: $3, 7, 12, 15, 19, 24, 30$?

a. 12

b. 15

c. 17

d. 19

23 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The mean of four numbers is 18. Three of the numbers are 12, 16, and 22. What is the fourth number?

a. 20

b. 22

c. 24

d. 26

24 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The heights of 5 basketball players in inches are 74, 76, 78, 80, and 82. What is the mean height?

a. 76 in

b. 78 in

c. 80 in

d. 82 in

25 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A scientist measured the weekly rainfall in millimeters over seven weeks: 2, 5, 8, 11, 14, 17, and 20. What was the median weekly rainfall?

a. 8

b. 10

c. 11

d. 14

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26 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the set: $5, 5, 5, 5, 5$?

a. 0

b. 1

c. 5

d. 25

27 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mean of the numbers: $10, 20, 30, 40, 50, 60$?

a. 30

b. 35

c. 40

d. 45

28 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A data set has minimum value 24 and range 36. What is the maximum value?

a. 50

b. 56

c. 60

d. 64

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The table shows the number of pets owned by each student in a class. What is the mean number of pets per student?

mber of p	0	1	2	3	4
ber of stu	9	8	4	2	5

a. 8.4

b. 1.5

c. 5.6

d. 2

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30 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The table shows the number of pets owned by each student in a class. What is the mean number of pets per student?

mber of p	0	1	2	3	4
ber of stu	6	6	4	2	8

a. 5.2

b. 10.4

c. 4

d. 2

31 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A biologist recorded the weights of six laboratory mice in grams: 14, 28, 19, 45, 12, and 33. What is the range of these weights?

a. 27

b. 31

c. 33

d. 45

32 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of the following even-sized data set: $4, 9, 11, 15, 18, 24$?

a. 11

b. 12

c. 13

d. 15

33 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the range of the scores: $85, 92, 78, 95, 88$?

a. 14

b. 15

c. 17

d. 18

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34 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A frequency table shows the number of siblings reported by 20 students. The results are shown in the table. What is the mode of the number of siblings?

nber of Sibl	0	1	2	3
nber of Stud	4	8	6	2

a. 0

b. 1

c. 2

d. 8

35 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a class of 25 students, the median test score is 84. If 12 students scored above 84, how many students scored at or below 84?

a. 12

b. 13

c. 14

d. 15

36 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The temperature in degrees Celsius over 5 days was: $-2, 0, 4, 7, 11$. What was the median temperature?

a. 0°C

b. 2°C

c. 4°C

d. 7°C

37 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

If the mean of three numbers is 10, what is the sum of the three numbers?

a. 10

b. 20

c. 30

d. 40

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The numbers of books read by five students last month are: 6, 10, 14, 18, and 22. What is the mean number of books read?

a. 12

b. 14

c. 16

d. 18

39 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A teacher recorded the number of questions answered correctly by eight students: 4, 8, 8, 12, 15, 15, 15, and 20. What is the mode of these scores?

a. 8

b. 12

c. 15

d. 20

40 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A survey asked 150 students how many hours per week they read. The results are shown in the table. How many students read at least 6 hours per week?

ours per we	0–2	3–5	6–8	9+
mber of stud	35	60	40	15

a. 40

b. 50

c. 55

d. 60

41 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

If five quiz scores have a sum of 420, what is the mean score?

a. 80

b. 82

c. 84

d. 86

42 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The mean of five test scores is 82. If four of the scores are 80, 85, 75, and 90, what is the fifth score?

a. 78

b. 80

c. 82

d. 84

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43 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

Find the median of the data set: $18, 22, 14, 30, 26$.

a. 18

b. 20

c. 22

d. 26

44 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the mode of the values: $12, 15, 18, 15, 21, 15, 18$?

a. 12

b. 15

c. 18

d. 21

45 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

What is the median of: $50, 60, 70, 80$?

a. 60

b. 65

c. 70

d. 75

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